

Mathematics

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Contents

1	Groups	2
1.1	Magma	2
1.2	Group	2

Chapter 1

Groups

1.1 Magmas

Definition 1.1.1. A magma is a pair $(S, *)$ with S a set and

$$* : S \times S \rightarrow S$$

an internal composition law.

Definition 1.1.2. A magma $(S, *)$ is said to be associative if $\forall a, b, c \in S, (a * b) * c = a * (b * c)$.

Definition 1.1.3. A magma $(S, *)$ is said to be commutative if $\forall a, b \in S, a * b = b * a$.

Definition 1.1.4. An identity element of a given magma $(S, *)$ is an element $e \in S$, such that $\forall a \in S, e * a = a * e = a$.

Theorem 1.1.5. *If a magma $(S, *)$ has an identity element, then it is unique.*

Proof. If $e, e' \in S$ are identity elements for $*$, then $e = e * e' = e'$. □

Definition 1.1.6. Given a magma $(S, *)$ with an identity element $e \in S$. An element $a \in S$ is said to be invertible if there exists b such that $a * b = e$.

1.2 Groups

Definition 1.2.1. A group is an associative magma with an identity element, such that every element is invertible.